STAFF REVIEW · H446 136 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Do Now — source answers

1. A junction/associative table.

2. A key made from two or more attributes together.

3. On a match or after examining all items.

4. O(log n) for sorted random-access data.

I Do — source worked steps

1. StudentID+CourseID key — Find the given data and name the required result.

2. Names partly depend on key — Explain why this intermediate stage is necessary.

3. Split Student and Course — StudentName depends only on StudentID; CourseName only on CourseID. Split into Student(StudentID PK,StudentName), Course(CourseID PK,CourseName), Enrolment(StudentID FK,CourseID FK,Grade) with composite PK.

We Do — guided CHECK

1. Composite OrderID+ProductID
Find the given data and name the required result.
ProductName depends on ProductID
Explain why this intermediate stage is necessary.
Move name to Product
ProductName depends on ProductID alone, not whole OrderID+ProductID key. Create Product(ProductID PK,ProductName) and retain OrderLine(OrderID,ProductID,Qty).

You Do — source mark schemes

1. Atomic field requirement [2], problem identified [2], separate rows or related table [2].

2. Composite key identified [2], description depends on ProductID only [2], violates 2NF and should move [2].

3. Student table [2], Club table [2], Membership table with composite key/date [2], correct FK/dependency reasoning [2].

Exit Ticket — source answer

1. False. Partial-key dependency requires a composite candidate key.